

Chapter 15

Chemical Equilibrium

15.1 The Concept of Equilibrium

15.2 The Equilibrium Constant

15.3 Understanding and Working with equilibrium
Constants

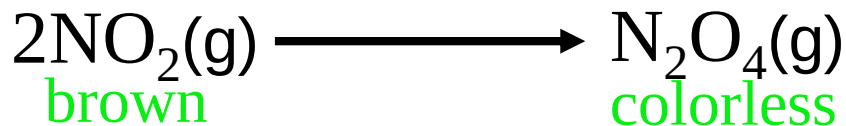
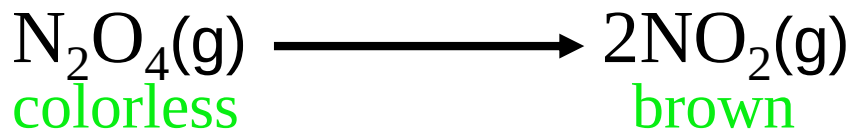
15.4 Heterogeneous Equilibrium

15.5 Calculating Equilibrium Constants

15.6 Applications of Equilibrium Constants

15.7 Le Chateliers Principle

15.1 The Concept of Equilibrium



Chemical
Equilibrium

- Reaction (2) is the reverse of reaction (1).

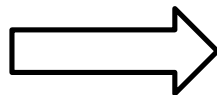


two different substances
with the same physical state

- Chemical equilibrium occurs when the forward and reverse reactions are proceeding at the same rate.
- Concentrations of reactants and products do not change with time once equilibrium is established.

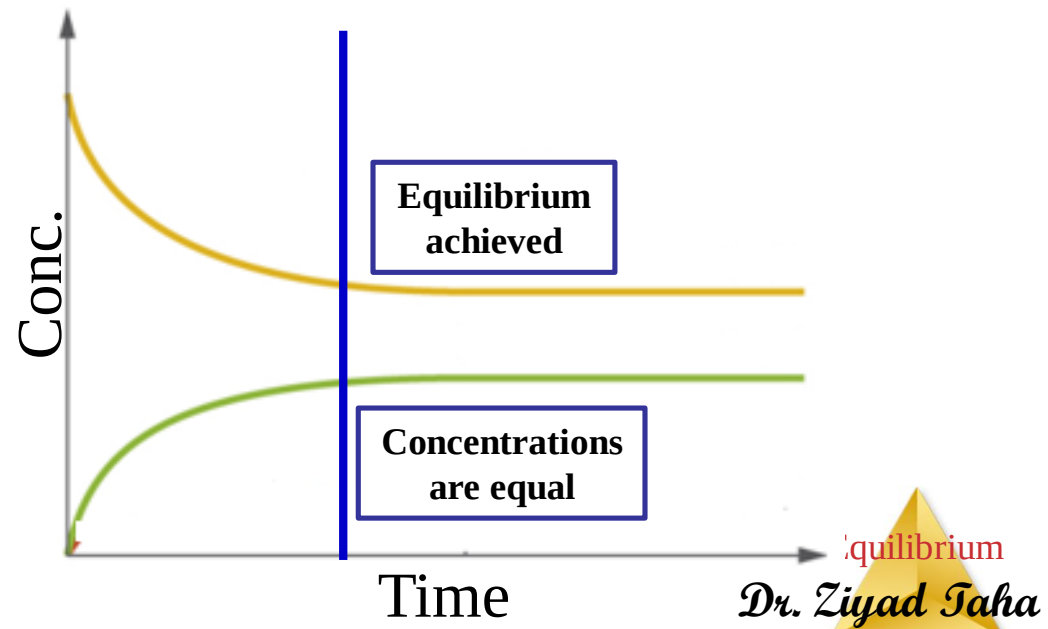
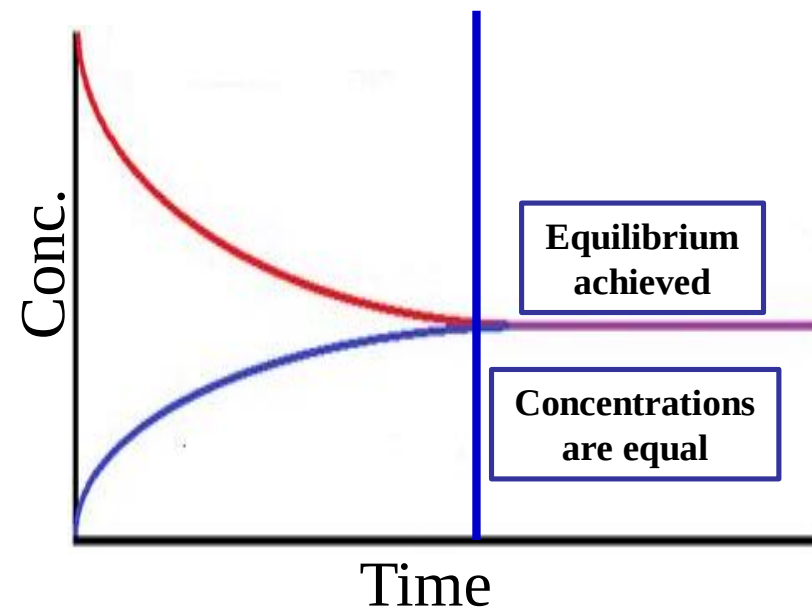
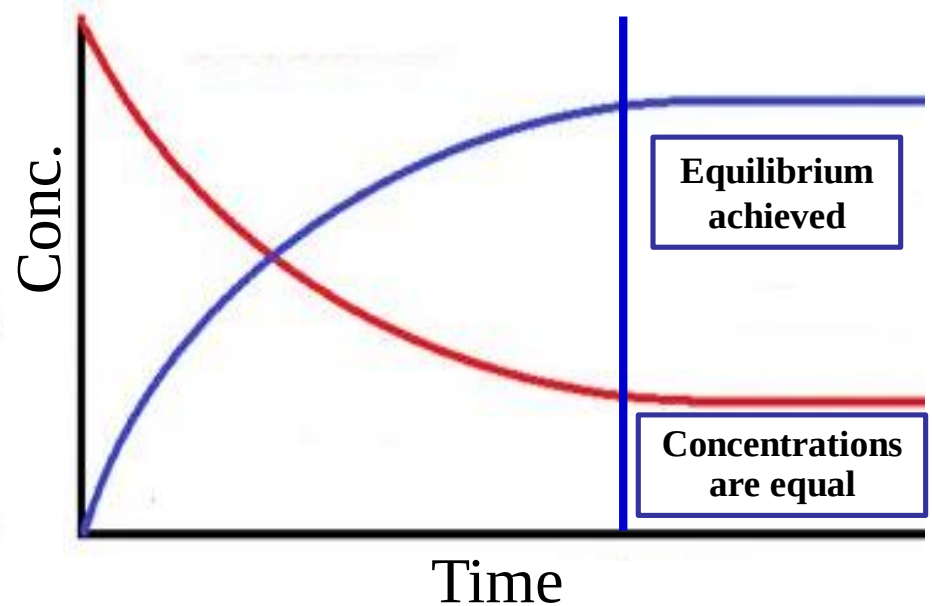
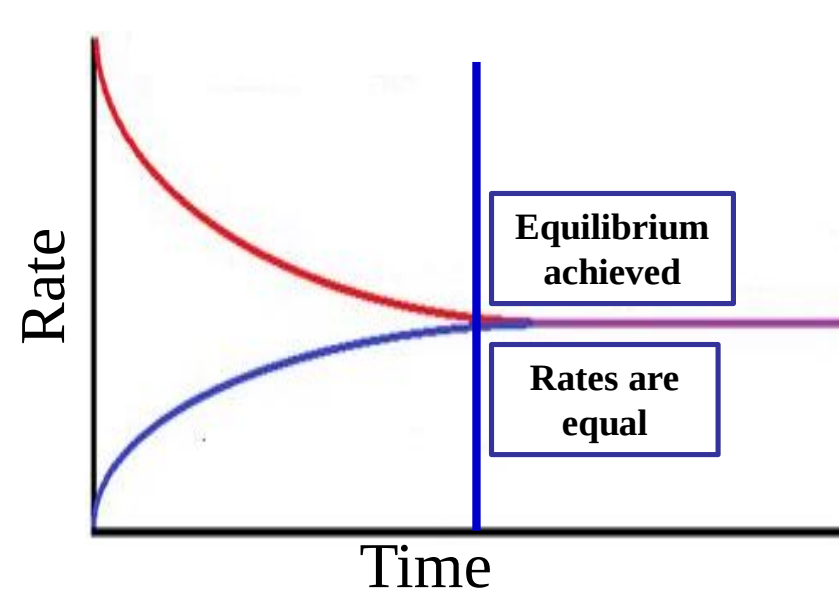


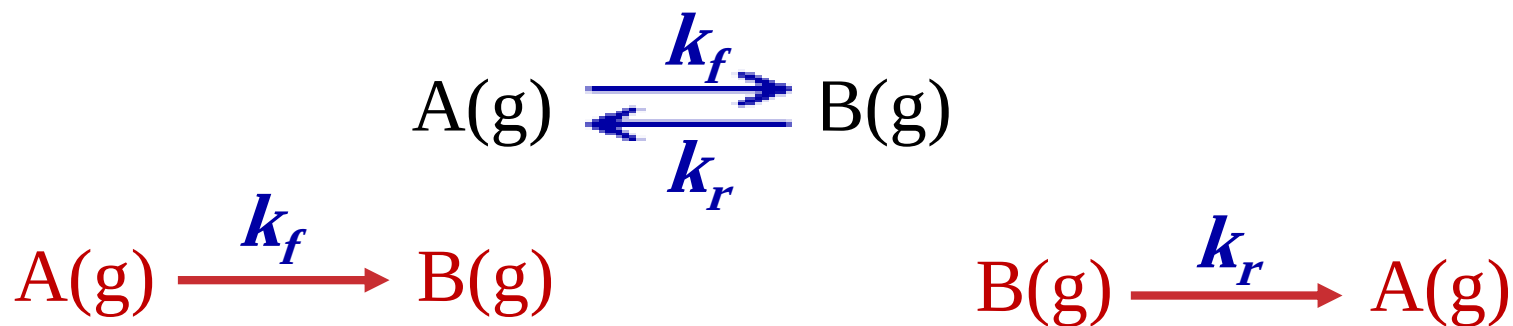
two phases of the
same substance



Physical
Equilibrium

A System at Equilibrium





■ At equilibrium: $\text{Rate}_f = \text{Rate}_r$

$$\text{Rate}_f = k_f[A]$$

$$\text{Rate}_r = k_r[B]$$

$$k_f[A] = k_r[B]$$

$$\boxed{K_c = \frac{[B]}{[A]}} \leftarrow \frac{k_f}{k_r} = \frac{[B]}{[A]} \rightarrow \boxed{\frac{k_f}{k_r} = K_c}$$

→ Equilibrium constant expression

15.2 The Equilibrium Constant

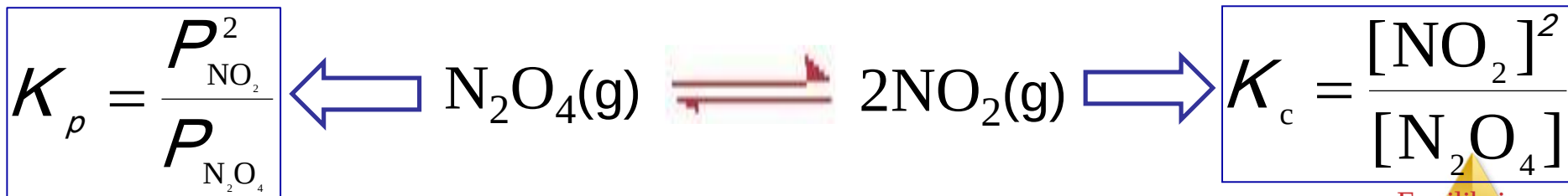
- Consider the following equilibrium system:



The Equilibrium Constant Expression

$$K_c = \frac{[D]^d [E]^e}{[A]^a [B]^b}$$

- $[D]$, $[E]$, $[A]$, and $[B]$ are the concentrations of reactants and products that exist at equilibrium.
- d , e , a , and b are the stoichiometric coefficients from the balanced chemical equation.



What is the relationship between K_p and K_c ???

Equilibrium

Dr. Ziyad Jaha

Relationship between K_c and K_p

- From the ideal gas law we know that

$$PV = nRT \Rightarrow \frac{n}{V} = \frac{P}{RT} \Rightarrow [] = \frac{P}{RT}$$



$$K_c = \frac{[\text{NO}_2]^2}{[\text{N}_2\text{O}_4]} \Rightarrow K_c = \frac{\left(\frac{P_{\text{NO}_2}}{RT}\right)^2}{\left(\frac{P_{\text{N}_2\text{O}_4}}{RT}\right)} = K_p (RT)^{-1}$$

$$K_p = K_c (RT)^1$$

- In general

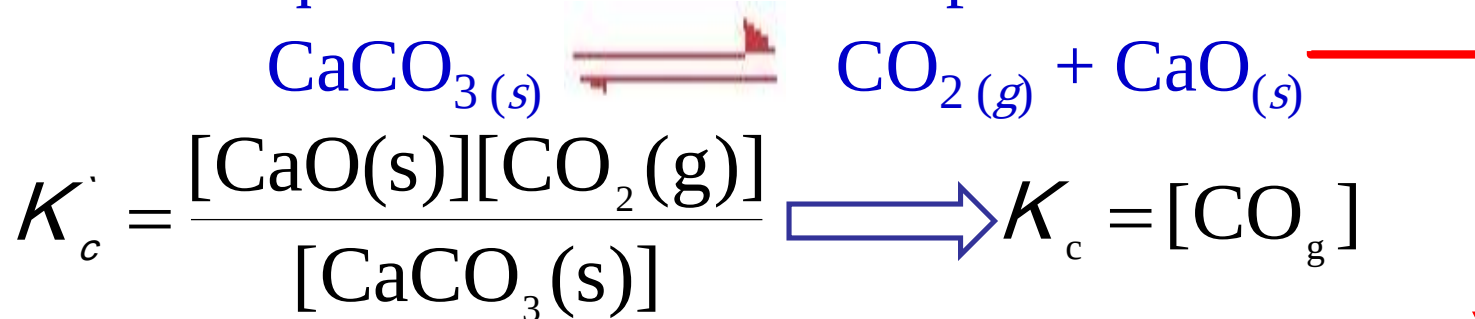
$$K_p = K_c (RT)^{\Delta n}$$

where: $\Delta n_{\text{gas}} = \sum (n_{\text{gas}})_{\text{products}} - \sum (n_{\text{gas}})_{\text{reactants}}$

Equilibrium

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Write the equilibrium constant expression for:



reactants and products that are
in different phases

Heterogeneous
equilibrium

$$K_p = P_{\text{CO}_2}; \quad K_p = K_c(RT)$$



$$K_c = \frac{[\text{H}_3\text{O}^+(\text{aq})][\text{CH}_3\text{CO}_2^-(\text{aq})]}{[\text{H}_2\text{O}(\text{l})][\text{CH}_3\text{CO}_2\text{H}(\text{aq})]}$$

$$K_c = \frac{[\text{H}_3\text{O}^+(\text{aq})][\text{CH}_3\text{CO}_2^-(\text{aq})]}{[\text{CH}_3\text{CO}_2\text{H}(\text{aq})]}$$

Homogeneous
equilibrium

reactants & products are all in the same phase

Sample Exercise

For the equilibrium: $\text{N}_2(\text{g}) + 3 \text{H}_2(\text{g}) \rightleftharpoons 2 \text{NH}_3(\text{g})$
 $K_c = 9.60$ at 300°C .

A. Write the equilibrium constant expression in term of concentration.

$$K_c = \frac{[\text{NH}_3]^2}{[\text{N}_2][\text{H}_2]^3}$$

Homogeneous
equilibrium

B. Write the equilibrium constant expression in term of partial pressure.

$$K_p = \frac{P_{\text{NH}_3}^2}{P_{\text{N}_2} P_{\text{H}_2}^3}$$

C. Calculate K_p for this reaction at this temperature

$$\Delta n = \{2\} - \{3 + 1\} = -2$$

$$K_p = K_c (RT)^{\Delta n} \implies K_p = 9.60(0.0821 \times (300 + 273))^{-2}$$

$$K_p = 4.34 \times 10^{-3}$$

Manipulating Equilibrium Constants

The equilibrium constant for $\text{N}_2\text{O}_4 (g) \rightleftharpoons 2 \text{NO}_2 (g)$ is 0.212. Calculate K_c for the $2 \text{NO}_2 (g) \rightleftharpoons \text{N}_2\text{O}_4 (g)$

$$\text{N}_2\text{O}_4 (g) \rightleftharpoons 2 \text{NO}_2 (g) \quad K_{c1} = \frac{[\text{NO}_2]^2}{[\text{N}_2\text{O}_4]} = 0.212$$

$$2 \text{NO}_2 (g) \rightleftharpoons \text{N}_2\text{O}_4 (g) \quad K_{c2} = \frac{[\text{N}_2\text{O}_4]}{[\text{NO}_2]^2} = \frac{1}{K_{c1}}$$
$$K_{c2} = \frac{1}{0.212} = 4.72$$

- The equilibrium constant of a reaction in the reverse reaction is the reciprocal of the equilibrium constant of the forward reaction.

$$K_2 = \frac{1}{K_1}$$

Manipulating Equilibrium Constants

The equilibrium constant for $\text{N}_2\text{O}_4(g) \rightleftharpoons 2 \text{NO}_2(g)$ is 0.212. Calculate K_c for the $2 \text{N}_2\text{O}_4(g) \rightleftharpoons 4 \text{NO}_2(g)$

$$\text{N}_2\text{O}_4(g) \rightleftharpoons 2 \text{NO}_2(g) \quad K_{c1} = \frac{[\text{NO}_2]^2}{[\text{N}_2\text{O}_4]} = 0.212$$

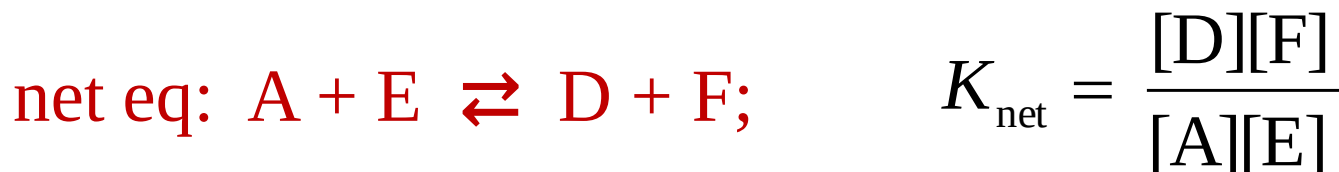
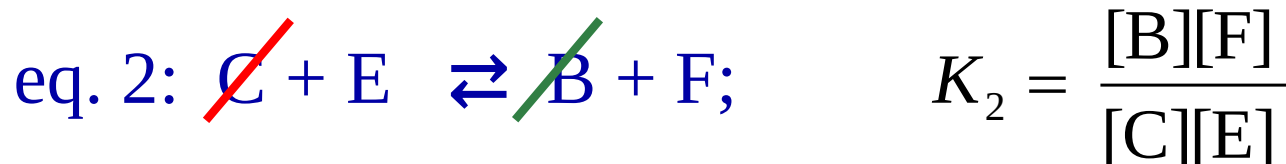
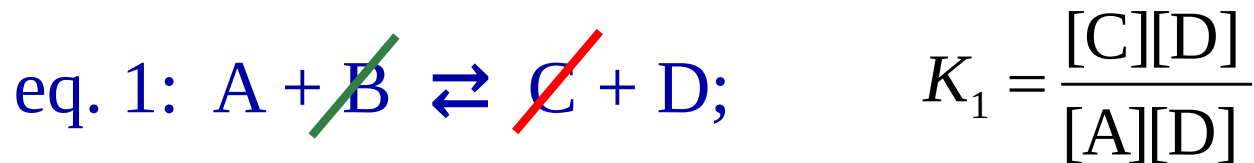
$$2 \text{N}_2\text{O}_4(g) \rightleftharpoons 4 \text{NO}_2(g) \quad K_{c2} = \frac{[\text{NO}_2]^4}{[\text{N}_2\text{O}_4]^2}$$

$$K_{c2} = \left(\frac{[\text{NO}_2]^2}{[\text{N}_2\text{O}_4]} \right)^2 = \left(K_{c1} \right)^2 = \left(0.212 \right)^2 = 0.0449$$

- If a chemical reaction multiply by a factor **n**, the equilibrium constant of the reaction raise to power **n**.

$$K_2 = (K_1)^n$$

Multiple Equilibrium



$$K_1 * K_2 = \frac{\cancel{[C]}[D]}{[A]\cancel{[B]}} * \frac{\cancel{[B]}[F]}{\cancel{[C]}[E]} \Rightarrow K_1 * K_2 = K_{\text{net}}$$

$K_1 * K_2 = K_{\text{net}}$

- If two or more reactions are added, the new K is the product of the individual K values.

$$K_{\text{net}} = K_1 * K_2 * K_3 \dots \text{etc}$$

Exercise

1. For the reaction: $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$, $K_c = 0.0602$ at 500°C . What is the equilibrium constant for the following reaction? $\text{NH}_3(\text{g}) \rightleftharpoons \frac{1}{2}\text{N}_2(\text{g}) + \frac{3}{2}\text{H}_2(\text{g})$

Answer: $K = 4.08$

2. For the reaction: $\text{SO}_3(\text{g}) \rightleftharpoons \text{SO}_2(\text{g}) + \frac{1}{2}\text{O}_2(\text{g})$. $K_c = 0.060$ at 1000 K . What is the equilibrium constant for the following reaction? $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$

Answer: $K = 278$

3. If $\text{N}_2(\text{g}) + \frac{1}{2}\text{O}_2(\text{g}) \rightleftharpoons \text{N}_2\text{O}(\text{g}); \quad K_{c1} = 2.4 \times 10^{-18}$
and $\text{N}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{NO}(\text{g}); \quad K_{c2} = 4.1 \times 10^{-31}$
What is the equilibrium constant for the reaction?



Answer: $K = 1.7 \times 10^{-13}$

Exercise

If: $\text{HF(aq)} \rightleftharpoons \text{H}^{\text{+}}(\text{aq}) + \text{F}^{-}(\text{aq}); \quad K_{\text{c1}} = 6.8 \times 10^{-4}$
and $\text{H}_2\text{C}_2\text{O}_4(\text{aq}) \rightleftharpoons 2 \text{H}^{\text{+}}(\text{aq}) + \text{C}_2\text{O}_4^{2-}(\text{aq}); \quad K_{\text{c2}} = 3.8 \times 10^{-8}$

What is the equilibrium constant for the reaction?



Answer: $K = 0.12$

Sample Exercise

1.0 mole of $\text{H}_2(\text{g})$ and 1.0 mole of $\text{I}_2(\text{g})$ are introduced into a 5.0 L sealed flask. The following reaction occurs until equilibrium is established.



At equilibrium, the mixture is found to contain 1.58 mole of HI.

- (a) What are the concentrations of H_2 , I_2 and HI at equilibrium?
- (b) Calculate the equilibrium constant K_c .



Initial # mol: 1.00 1.00 0.000

Volume, L: 5.00 5.00 5.00

Initial [], M: 0.200 0.200 0.000

Change in [], M: - x - x + 2x

Equilibrium [], M 0.20 - x 0.20 - x 2x

Equilibrium [], M 0.042 0.042 0.316

$$x = 0.158 \quad \leftarrow 2x = 0.316$$

$$K_c = \frac{[\text{HI}]^2}{[\text{H}_2][\text{I}_2]} = \frac{(0.316)^2}{(0.042)^2} \Rightarrow K_c = 57$$

Reactants

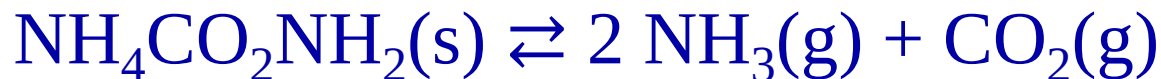


Products

$K \gg 1$, the reaction is product-favored; product predominates at equilibrium.

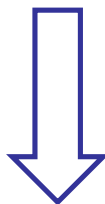
Sample Exercise

Consider the following reaction:

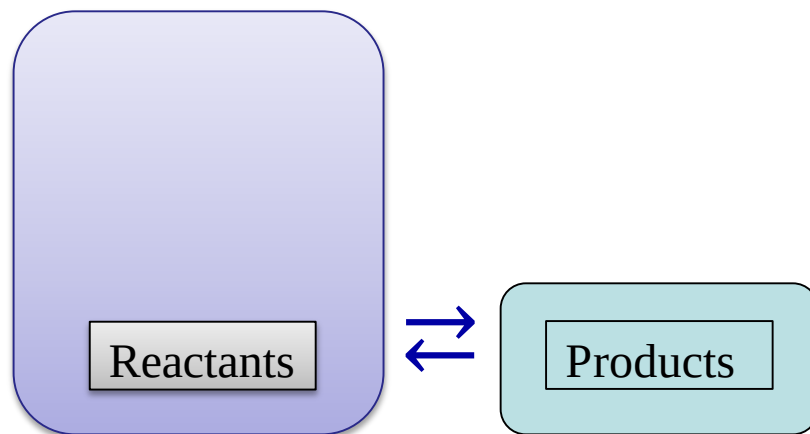


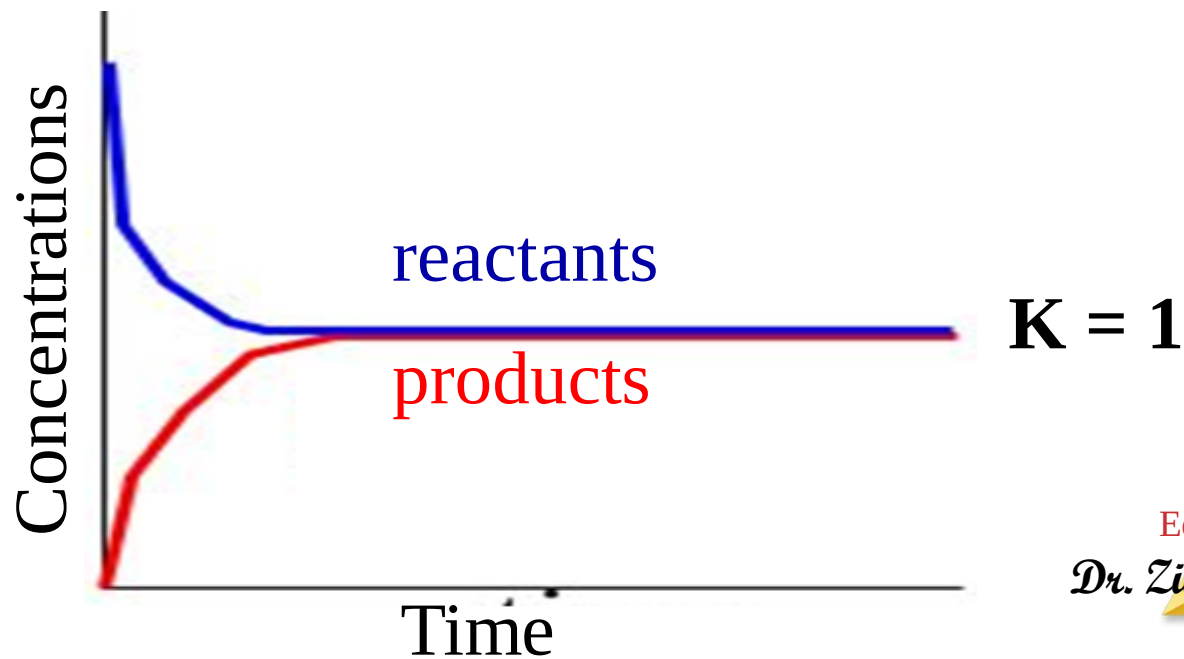
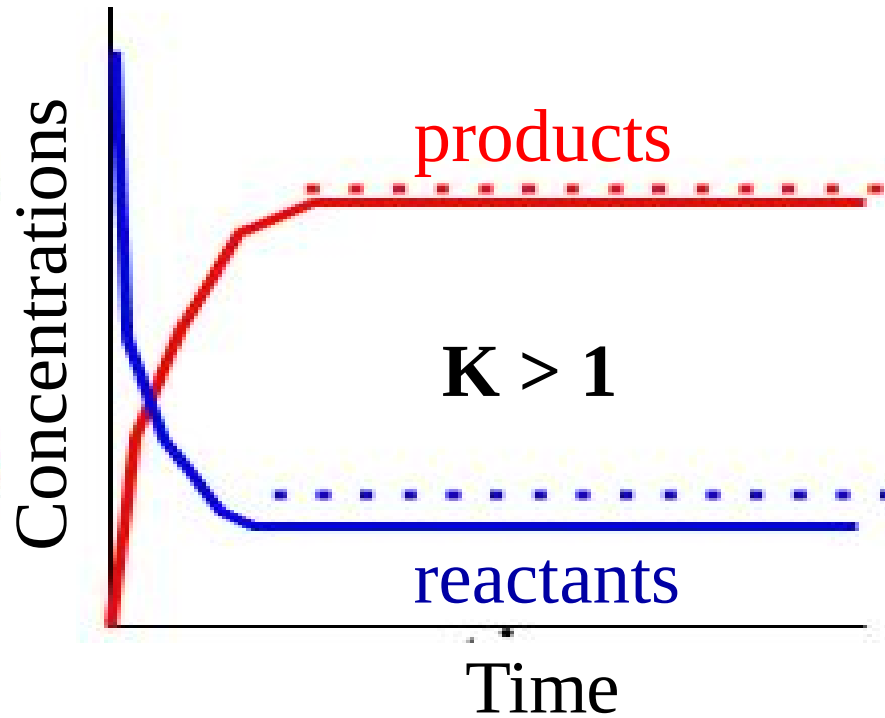
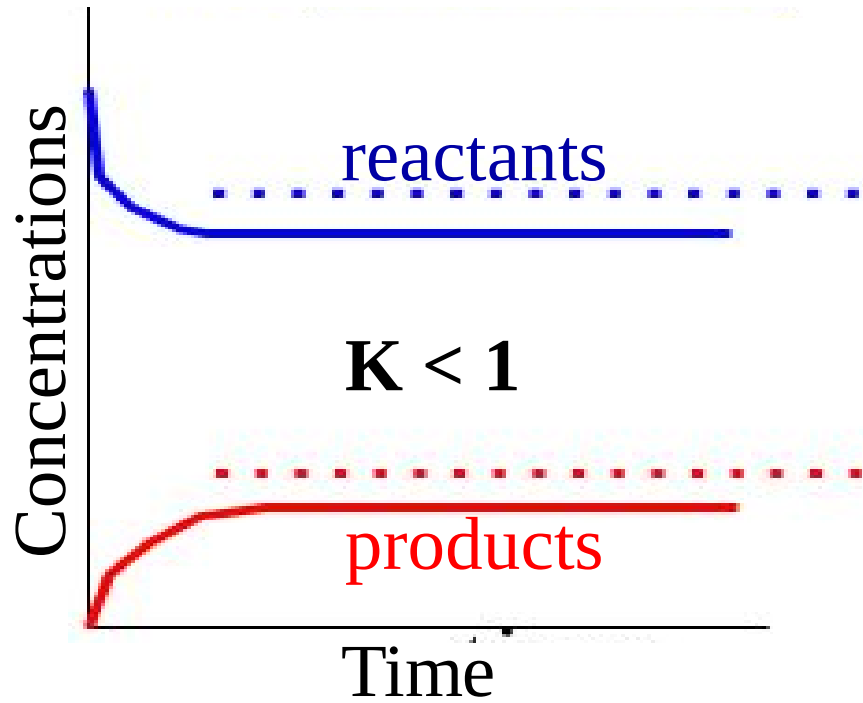
It was found that at 40°C, the total gas pressure is 0.363 atm, calculate K_p and K_c ?

Answer: $K_p = 7.08 \times 10^{-3}$ and $K_c = 1.07 \times 10^{-5}$



If $K \ll 1$, the reaction is reactant-favored; reactant predominates at equilibrium.





Sample Exercise

In an experiment, 1.0 mol each of CO and H₂ were allowed to react in a sealed 10.0-L reaction vessel at 500 K. When the equilibrium was established, the mixture was found to contain 0.0892 mole of CH₃OH. (R = 0.0821 L.atm/Mol.K)



A) What are the equilibrium concentrations of CO, H₂ and CH₃OH?

Answer: [CO] = 0.0911 M; [H₂] = 0.0822 M; [CH₃OH] = 0.00892 M

B) Calculate the equilibrium constants K_c and K_p for this reaction at 500 K?

Answer: $K_c = 14.5$; $K_p = 8.60 \times 10^{-3}$

Sample Exercise

0.50 mol of HI is introduced into a 1.0 L sealed flask. HI decomposes to produce H_2 and I_2 until an equilibrium is established. An analysis of the equilibrium mixture shows that 0.105 mole of HI has decomposed. Calculate the equilibrium concentrations of H_2 , I_2 and HI, and the equilibrium constant K_c for the following reaction:

	$\text{H}_2(\text{g})$	+	$\text{I}_2(\text{g})$	$\rightleftharpoons$	$2\text{HI}(\text{g})$
Initial # mol:	0.00		0.00		0.50
Volume, L:	1.00		1.00		1.00
Initial [], M:	0.000		0.000		0.50
Change in [], M:	+ x		+ x		- 2x
Change in [], M:	+0.0525		+ 0.0525		- 0.105
Equilibrium [], M	0.0525		0.0525		0.395

$$K_c = 56.6$$

Sample Exercise

0.50 mol of HI is introduced into a 1.0 L sealed flask. HI decomposes to produce H_2 and I_2 until an equilibrium is established. An analysis of the equilibrium mixture shows that 0.105 mole of HI has decomposed. Calculate the equilibrium concentrations of H_2 , I_2 and HI, and the equilibrium constant K_c for the following reaction:

	$\text{H}_2(\text{g})$	+	$\text{I}_2(\text{g})$	$\rightleftharpoons$	$2\text{HI}(\text{g})$
Initial # mol:	0.00		0.00		0.50
Volume, L:	1.00		1.00		1.00
Initial [], M:	0.000		0.000		0.50
Change in [], M:	+ x		+ x		- 2x
Change in [], M:	+0.0525		+ 0.0525		- 0.105
Equilibrium [], M	0.0525		0.0525		0.395

$K_c = 56.6$

Sample Exercise

A flask is charged with 2.00 atm of $\text{NO}_2(\text{g})$ and 1.00 atm of $\text{N}_2\text{O}_4(\text{g})$ at 25 °C and allowed to reach equilibrium. When equilibrium is established, the partial pressure of NO_2 has decreased by 1.24 atm.

(a) What are the partial pressures of NO_2 and N_2O_4 at equilibrium?

(b) Calculate K_p and K_c for following reaction at 25 °C.



Answer: $K_p = 2.80$; $K_c = 68.6$

15.6 Applications of equilibrium constants

Predicting the direction of a reaction

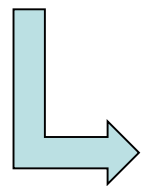
For the reaction: $\text{CO(g)} + 2\text{H}_2\text{(g)} \rightleftharpoons \text{CH}_3\text{OH(g)}$

$K_c = 15.0$ at a certain temperature. Is a reaction mixture that contains 0.40 M CO , 0.80 M H_2 , and $0.10 \text{ M CH}_3\text{OH}$ at equilibrium? If not, in which direction will the net reaction occur? What will be their concentrations when equilibrium is established?



Initial [], M: 0.40 0.80 0.10

$$Q = \frac{[\text{CH}_3\text{OH}]}{[\text{H}_2]^2[\text{CO}]} \Rightarrow Q = \frac{(0.10)}{(0.80)^2(0.40)} \Rightarrow Q = 0.39$$



reaction quotient



concentrations are not necessarily at equilibrium.

$Q_c < K_c \Rightarrow$ a net forward reaction $\Rightarrow$ From left to right

- For any system or reaction:
 1. Knowing the equilibrium constant, we can predict whether or not a reaction mixture is at equilibrium, and we can predict the direction of net reaction.

$Q_c = K_c \rightarrow$ the reaction is at equilibrium (no net forward reaction)

$Q_c > K_c \rightarrow$ the reaction is not at equilibrium and there's a net forward reaction

$Q_c < K_c \rightarrow$ the reaction is not at equilibrium and there's a net reverse reaction

Sample Exercise



initial: 0.40 0.80 0.10

change: - x - 2x + x

equilibrium: 0.40 - x 0.80 - 2x 0.10 + x

$$K_c = \frac{[\text{CH}_3\text{OH}]}{[\text{H}_2]^2[\text{CO}]} \Rightarrow 15 = \frac{(0.10 + x)}{(0.80 - 2x)^2(0.40 - x)}$$

■ Calculate the equilibrium concentrations

Sample Exercise

6.0 mol each of H_2 and I_2 are introduced into a 12.0 L sealed flask. The following reaction occurs until equilibrium is established. Calculate the equilibrium concentrations of H_2 , I_2 and HI .

	$\text{H}_2(\text{g})$	+	$\text{I}_2(\text{g})$	$\rightleftharpoons$	$2\text{HI}(\text{g})$	$K_c=50.5$
Initial # mol:	6.0		6.0		0.0	
Volume, L:	12.0		12.0		12.0	
Initial [], M:	0.50		0.50		0.00	
Change in [], M:	+ x		+ x		- 2X	
Equilibrium [], M	$0.50 - x$		$0.50 - x$		+ 2x	

$$K_c = \frac{[\text{HI}]^2}{[\text{I}_2][\text{H}_2]} \Rightarrow 50.5 = \frac{[2x]^2}{[0.50 - x][0.50 - x]}$$

$$\sqrt{50.5} = \frac{2x}{0.50 - x} \Rightarrow \boxed{x = 0.39}$$



Initial [], M: 0.50 0.50 0.00

at equilibrium [], M; 0.11 0.11 0.78

If the rxn proceeds to completion

0.0 0.0 1.0

- Calculate the percentage yield of formation of HI.

$$\% \text{Yield} = \frac{[\text{HI}]_{\text{eq}}}{[\text{HI}]_{\text{f}}} \times 100$$

is the [HI] if the rxn proceeds to completion.

$$\% \text{Yield} = \frac{[0.78]}{[1.0]} \times 100$$

$$\% \text{Yield} = 78\% \Rightarrow \boxed{\text{the reaction is product-favored}}$$

15.7 Le Châtelier's Principle

“If a system at equilibrium is disturbed by a change in temperature, pressure, or the concentration of one of the components, the system will shift its equilibrium position so as to counteract the effect of the disturbance.”

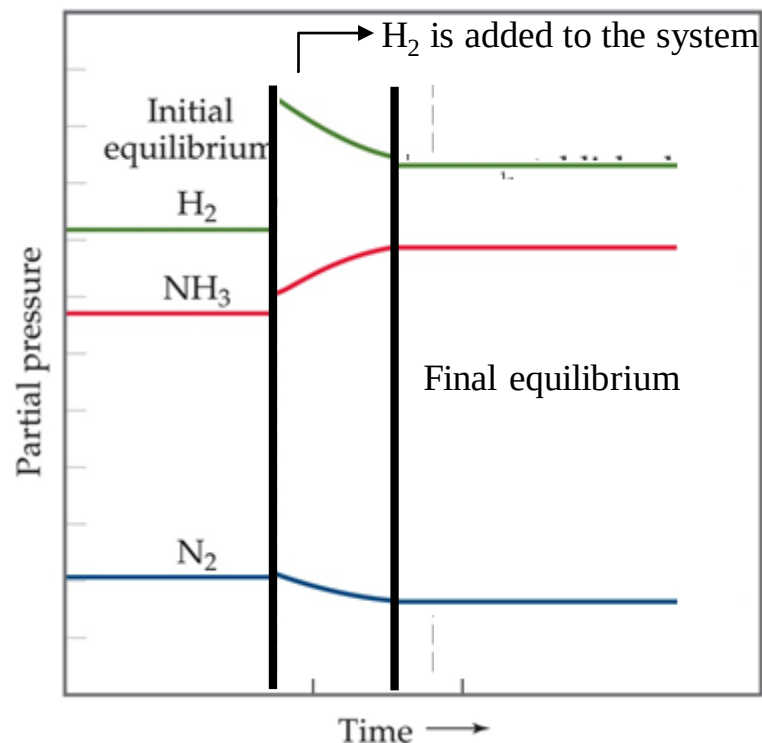


$$K_c = \frac{[\text{NH}_3]^2}{[\text{N}_2][\text{H}_2]^3}$$

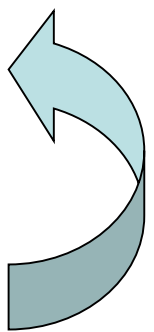
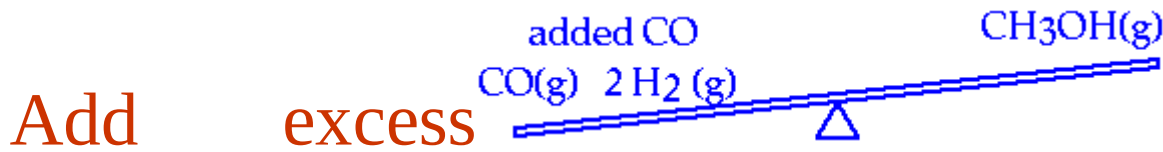
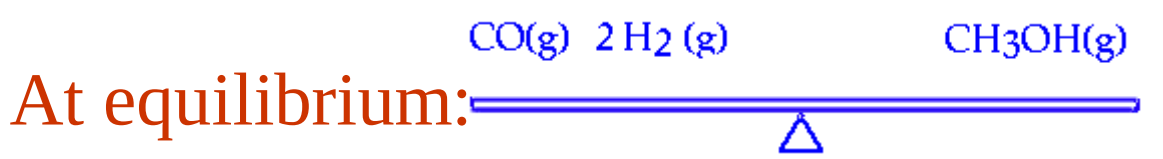
- If H_2 is added to the system
 $Q_c < K_c$ Shifts from left to right

N_2 will be consumed

The two reagents will form more NH_3 .



Consider this reaction:

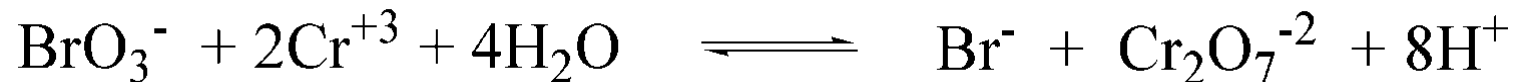


To return to equilibrium (balance), some (not all) CO and H₂ are converted to CH₃OH

If all added CO was converted to CH₃OH, then reaction would be unbalanced by the amount of product

Sample exercise:

Consider this reaction:



$$K = \frac{[\text{Br}^-][\text{Cr}_2\text{O}_7^{2-}][\text{H}^+]^8}{[\text{BrO}_3^-][\text{Cr}^{3+}]^2} = 1 \times 10^{11} \text{ at } 25^\circ\text{C}$$

At one equilibrium state:

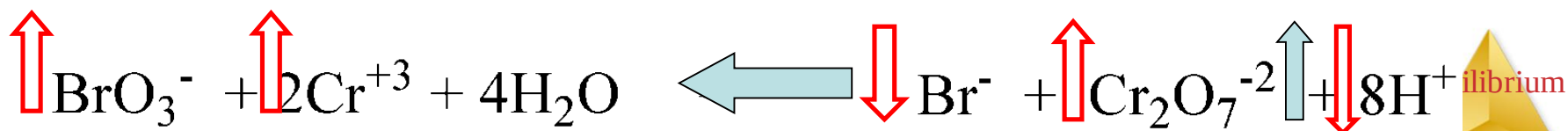
$$[\text{H}^+] = 5.0 \text{ M} \quad [\text{Cr}_2\text{O}_7^{2-}] = 0.10 \text{ M} \quad [\text{Cr}^{3+}] = 0.0030 \text{ M}$$

$$[\text{Br}^-] = 1.0 \text{ M} \quad [\text{BrO}_3^-] = 0.043 \text{ M}$$

What happens when:

$$[\text{Cr}_2\text{O}_7^{2-}] \text{ increased from } 0.10 \text{ M to } 0.20 \text{ M}$$

According to Le Châtelier's Principle, reaction should go back to left to off-set dichormate on right:

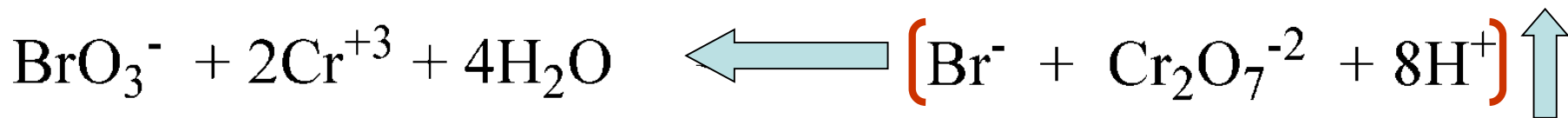


- Use reaction quotient (Q):

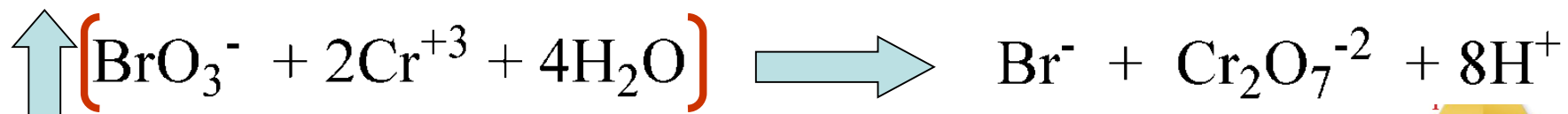
$$Q = \frac{[\text{Br}^-][\text{Cr}_2\text{O}_7^{2-}][\text{H}^+]^8}{[\text{BrO}_3^-][\text{Cr}^{3+}]^2} = \frac{(1.0)(0.20)(5.0)^8}{(0.043)(0.0030)^2} = 2 \times 10^{11} > K$$

- Because $Q > K$, the reaction must go to the left to decrease numerator and increase denominator. Continues until $Q = K$

- If products are added (or reactants are removed), the reaction goes to the left.



- If reactants are added (or products are removed), the reaction goes to the right.



Effects of pressure and volume changes

- Changes in pressure in aqueous solution, liquids and solids, has no effect on the chemical equilibrium.
- If the volume of a gas mixture is compressed, the overall gas pressure will increase.

If $\Delta n < 0$ From left to right

If $\Delta n > 0$ From right to left

If $\Delta n = 0$ No effect on the position of the equilibrium

- However, there will be no effect to equilibrium if the total gas pressure is increased by adding an inert gas that is not part of the equilibrium system.

Sample Exercise

- If pressure is increased by decreasing the volume

$$\Delta n = 2 - 3 = -1 < 0$$

From left to right (forwarded rxn)

Sample Exercise

- If mixture is compressed, pressure increases,

$$\Delta n = 2 - 1 = 1 > 0$$

From right to left (reverse rxn)

- If volume expands and pressure drops

$$\Delta n = 2 - 1 = 1 > 0$$

From left to right (forwarded rxn)

- If 1.0 atm of He(g) is added to the rxn vessel.

no effect on the equilibrium position, the total gas pressure is increased but the partial pressure does not.

Sample Exercise

$$\Delta n = 2 - 1 = 1 > 0$$

Equilibrium is not affected

- Reducing or increasing the volume will cause equal effect on both sides – no net reaction will occur.

Equilibrium

Dr. Ziyad Jaha

The Effect Temperature on Equilibrium

- Consider the following exothermic reaction:



- For exothermic reaction:

- When heat is added to increase temperature

Equilibrium shifts from right to left
(reverse rxn will occur)

K_c decreases

- If heat is removed to reduce temperature

Equilibrium shifts from left to right
(forward rxn will occur)

K_c increases

- Exothermic reactions favor low temperature conditions.

The Effect Temperature on Equilibrium

- Consider the following endothermic reaction:



- For endothermic reaction:

- When heat is added to increase temperature

Equilibrium shifts from left to right
(forward rxn will occur)

K_c increases

- If heat is removed to reduce temperature

Equilibrium shifts from right to left
(reverse rxn will occur)

K_c decreases

- Endothermic reactions favor high temperature conditions.

Sample Exercise

Determine whether the following reactions favor high or low pressures?



$\Delta n = 2 - 3 = -1 < 0 \implies$ high pressures



$\Delta n = 2 - 1 = 1 > 0 \implies$ low pressures



$\Delta n = 1 - 3 = -2 < 0 \implies$ high pressures



$\Delta n = 2 - 1 = 1 > 0 \implies$ low pressures



$\Delta n = 2 - 2 = 0 \implies$ Neither (no effect)

Sample Exercise

Determine whether the following reactions favors high or low temperature?



Exothermic $\Longrightarrow$ **Low temperature**



Exothermic $\Longrightarrow$ **Low temperature**



Exothermic $\Longrightarrow$ **Low temperature**

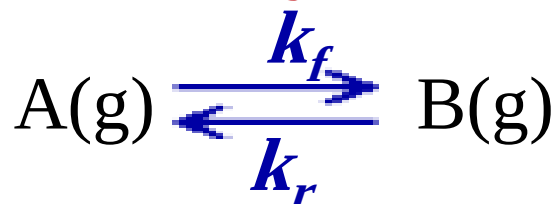


Endothermic $\Longrightarrow$ **High temperature**



Exothermic $\Longrightarrow$ **Low temperature**

The Effect Catalysts on Equilibrium



- At equilibrium: $\text{Rate}_f = \text{Rate}_r$

$$k_f[\text{A}] = k_r[\text{B}]$$

$$\frac{k_f}{k_r} = \frac{[\text{B}]}{[\text{A}]} \Rightarrow \frac{k_f}{k_r} = K_c$$

- Catalysts increase the rate of both the forward and reverse reactions equally.
- Equilibrium is achieved faster, but the equilibrium composition remains unaltered.

